



Movie Recommender System

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Project Motivation



Modern streaming platforms face information overload, making personalization essential



Users expect accurate, adaptive recommendations based on their preferences



Collaborative filtering provides a data-driven way to uncover user taste patterns



MovieLens offers a rich, clean benchmark for building realistic recommenders



IMDb ratings provide external validation of prediction quality

Datasets Used

Data Set	Characteristics	Purpose
MovieLens 100K	• 100,000 ratings • 610 users, 9,742 movies • Columns: userId, movieId, rating, timestamp	• Core training data for collaborative filtering models • Builds the user-item interaction matrix
Kaggle Movies Metadata	• ~45,000 movies • Title, genres, runtime, budget, revenue, release date • Scraped from TMDB	• Enriches movie attributes for EDA and interpretation • Merged with MovieLens via TMDB IDs
IMDb Ratings Dataset	• IMDb audience rating and vote count • Columns: tconst, averageRating, numVotes	• External validation benchmark • Used to measure correlation with predicted ratings

Pipeline Overview

1. Download & Load Data

Retrieve MovieLens, Kaggle, and IMDb data programmatically

2. Clean & Merge Metadata

Parse genres, derive release year, normalize IDs, and integrate all datasets

3. Exploratory Data Analysis (EDA)

Analyze rating pattern, language distribution, genres, and release year trends

4. Build Collaborative Filtering Dataset

Prepare Surprise Trainset and create train/test splits for model evaluation

5. Train & Tune Models

Tune SVD and KNNBaseline hyperparameters. Generate predictions

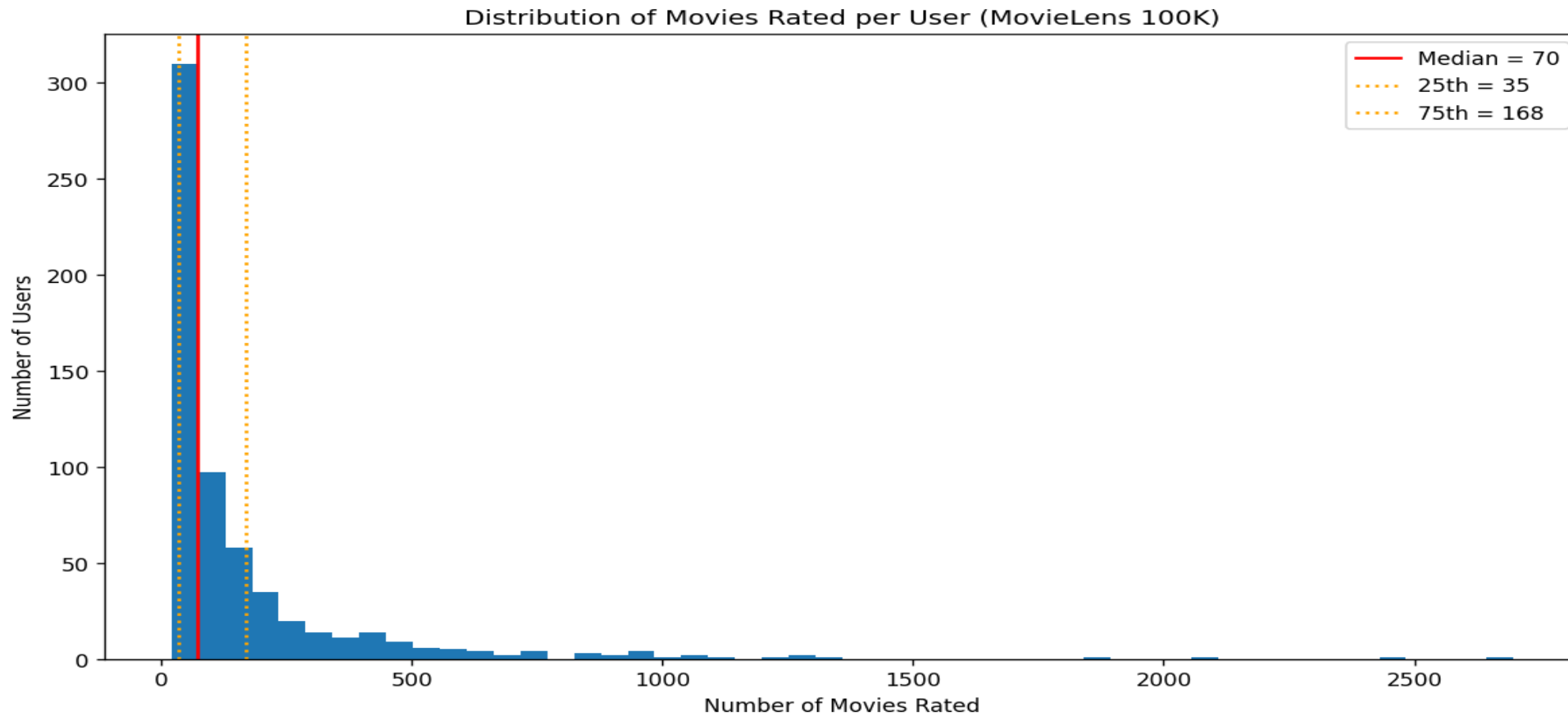
6. Generate Top-N Recommendations

Produce SVD and KNN recommendations for selected users

7. External Validation (IMDb)

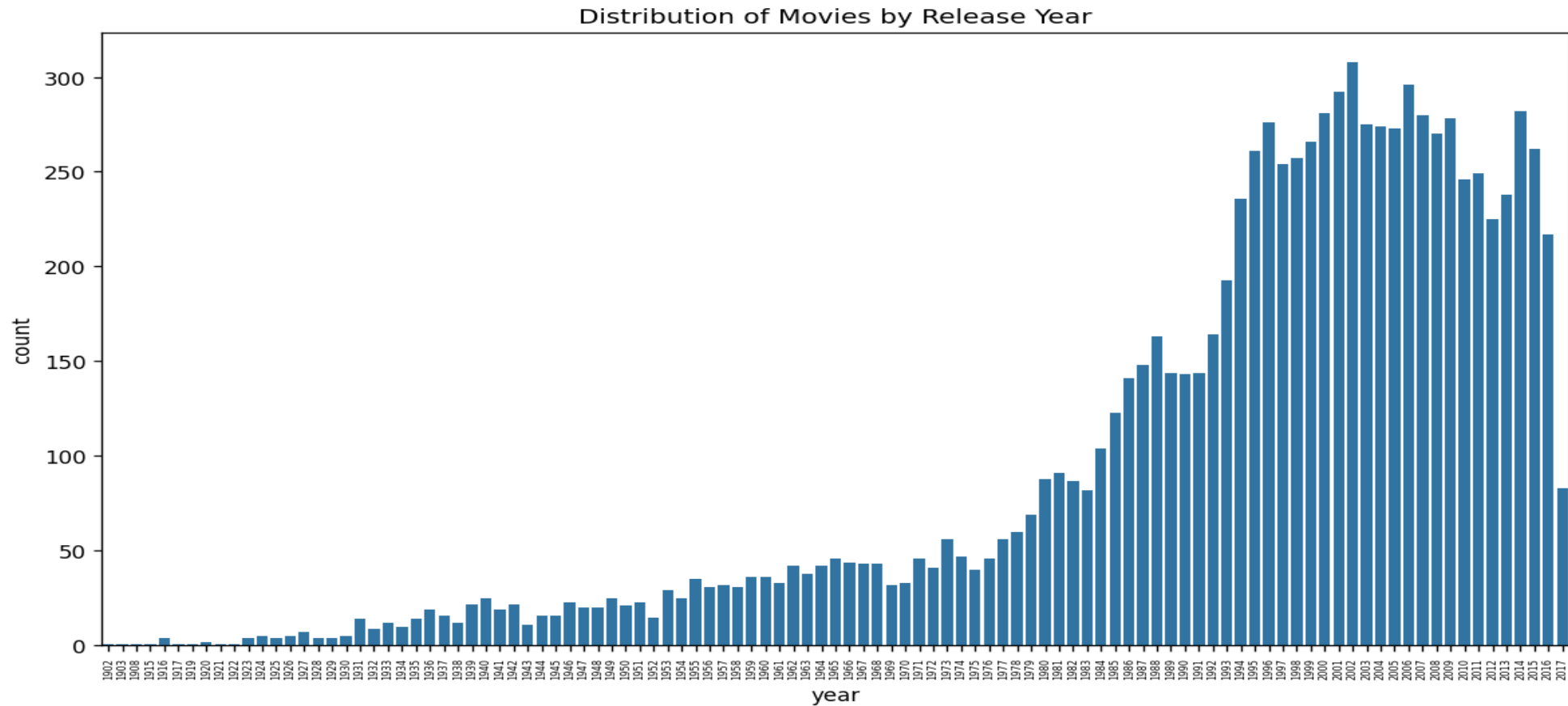
Compare predicted ratings with IMDb audience ratings via correlation

EDA – User Activity



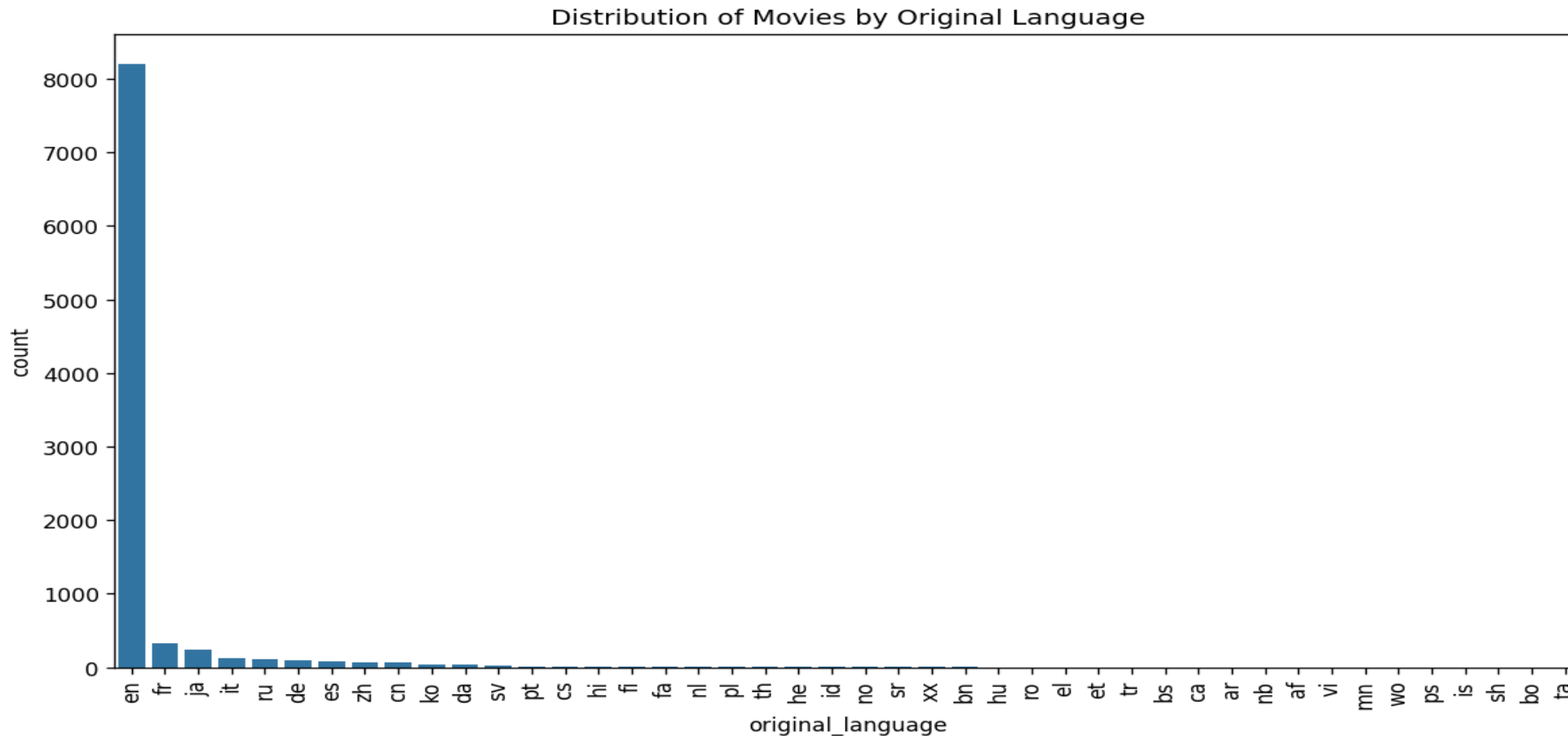
Most users rate fewer than 200 movies; distribution is long-tailed

EDA – Movies by Release Year



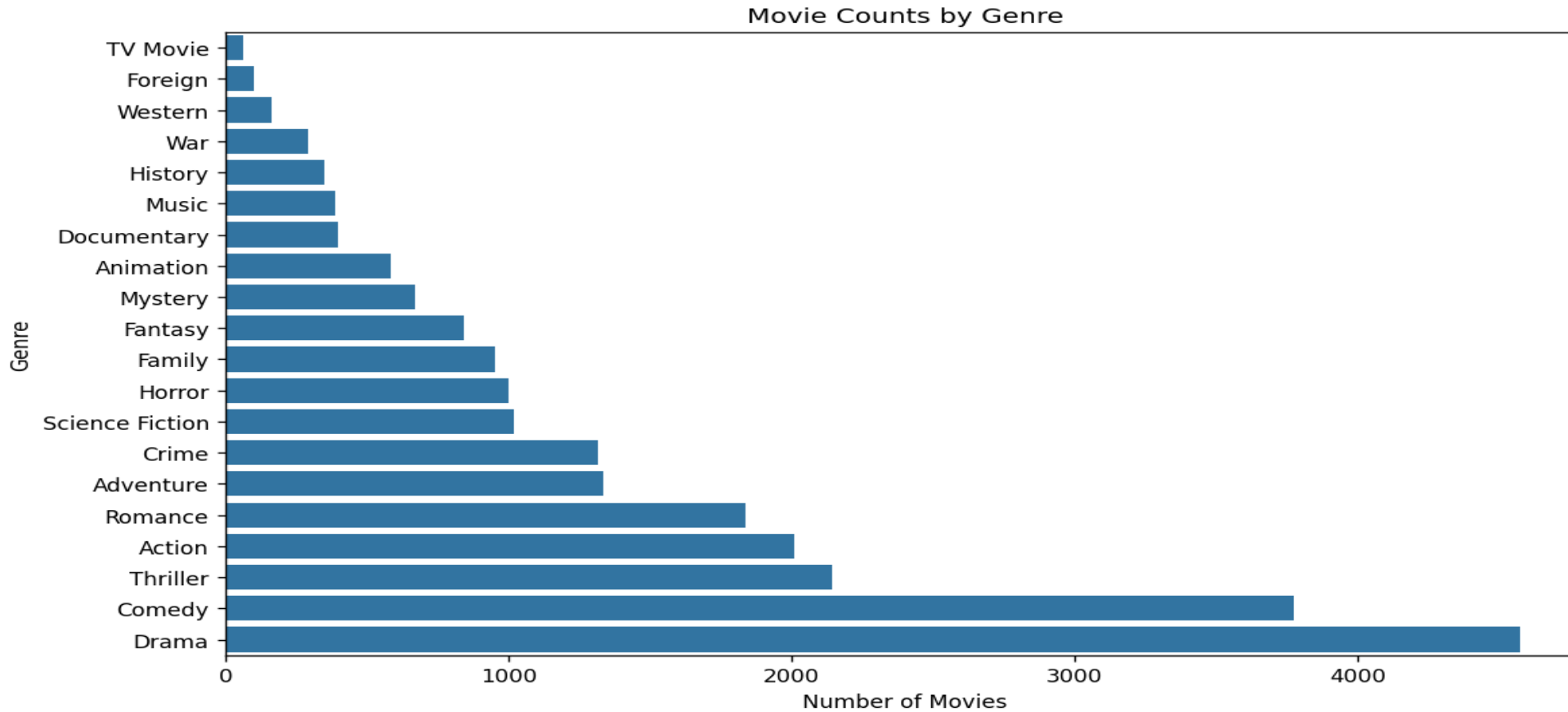
Movie count rises sharply after 1980; most movies from 1990's and 2000's

EDA – Movies by Language



Dataset dominated by English-language films; long tail of global languages

EDA – Movies by Genre



Drama, Comedy, and Thriller dominate; niche genres have far fewer movies

Models Used

SVD

- Learns latent features to model user preferences and movie traits
- Best parameters (GridSearchCV):
 - n_factors: 300
 - reg_all: 0.1
 - lr_all: 0.025

KNNBaseline

- Computes similarity between items while adjusting for user and item rating biases using baseline estimates
- Best parameters (GridSearchCV):
 - k: 50
 - sim_options:
{“name”:“pearson_baseline”,
“user_based”:False}

Model Performance

- SVD achieves the lowest error on both CV and test sets, showing better generalization
- KNNBaseline performs competitively but shows slightly higher variance and weaker predictive precision
- The small gap in CV and Test RMSE suggests both models are well-fitted

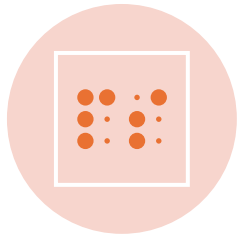
Prediction Error		
Model	CV RMSE	Test RMSE
SVD	~ 0.86	~ 0.858
KNNBaseline	~ 0.863	~ 0.86

External Validation (IMDB)

- SVD's predictions show a stronger alignment with IMDB audience ratings
- KNNBaseline still captures meaningful structure, but its correlation is lower, indicating weaker external generalization
- Overall, the external validation supports SVD as the more reliable model for reflecting real-world movie quality

Model	Correlation
SVD	0.612
KNNBaseline	0.536

Challenges Faced



MINOR METADATA CLEANING

Occasional
missing dates,
languages, and
genres



DATASET MERGING

Combining
movielens ratings
with TMDb/imdb
metadata
smoothly



SURPRISE FORMATTING

Converting data
into surprise's
dataset rather
than dataframes



GRIDSEARCHCV BEHAVIOR

Surprise's tuning
API requires the
full dataset,
unlike scikit-learn



LONG TUNING CYCLES

Selecting
hyperparameter
ranges and slow
CV runs



Thank You
